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Computational Modelling in the Humanities and Social Sciences

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Today

Modelling text

1. Representation of text
   - Unicode and UTF-8
2. Text preprocessing
3. Representations of text for Natural Language Processing

Motivation:

- Vast amounts of data exist (only) as unstructured text
- Structured data is more easily processed, but expensive to create

Modelling text

```
example_text = "This is an example"
```

```
This is an example
```

Character indexes	84	104	105	115	32	105	115	32	97	110	32	97	110	97	109	112	108	97

Motivation: same sequence of numbers => same literal content

```
example_text[2:4] == example_text[5:7]
```

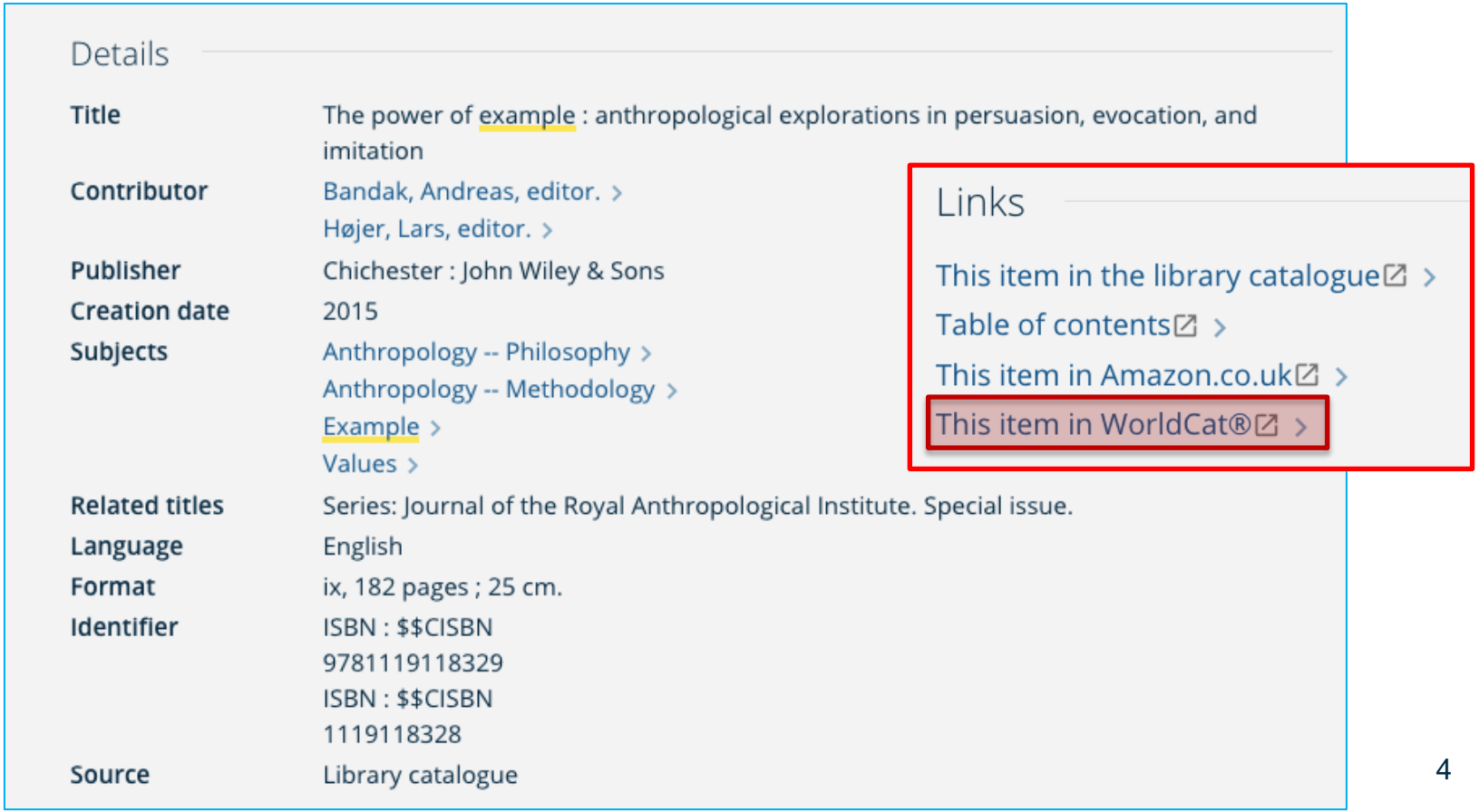
- Q: Where do the numbers come from?
  - What do they represent?
- A: Some *vocabulary* tells us which to use



Encodings (according to Python 3.7) which will / will not represent "typical" (i.e. English) strings such as the one above with *exactly* the same sequence of numbers

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Real-world example: Durham University Library



Details

Title	The power of example : anthropological explorations in persuasion, evocation, and imitation
Contributor	Bandak, Andreas, editor. › Højer, Lars, editor. ›
Publisher	Chichester : John Wiley & Sons
Creation date	2015
Subjects	Anthropology -- Philosophy › Anthropology -- Methodology › Example › Values ›
Related titles	Series: Journal of the Royal Anthropological Institute. Special issue.
Language	English
Format	ix, 182 pages ; 25 cm.
Identifier	ISBN : $$CISBN 9781119118329 ISBN : $$CISBN 1119118328
Source	Library catalogue

Links

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Author

Andreas Bandak

Lars Højer (1)

Year

2015

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1. 

The Power of Example: Anthropological Explorations in Persuasion, Evocation.
by Andreas Bandak; Lars HÓjer
Print book
Language: English
Publisher: John Wiley & Sons, 2015.

2. 

The power of example : anthropological explorations in persuasion, evocation, and imitation
by Andreas Bandak; Lars Højer;
Print book View all formats and languages »
Language: English
Publisher: Chichester : John Wiley & Sons, 2015.

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WorldCat

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HTML source

```
<link rel="meta" type="application/rdf+xml" href="http://www.worldcat.org/oclc/942288255.rdf" />
<link rel="meta" type="application/ld+json" href="http://www.worldcat.org/oclc/942288255.jsonld" />
<link rel="meta" type="text/turtle" href="http://www.worldcat.org/oclc/942288255.ttl" />
<link rel="meta" type="text/plain" href="http://www.worldcat.org/oclc/942288255.nt" />
<meta name="keywords" content="" />
```

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```
<link rel="meta" type="application/rdf+xml" href="http://www.worldcat.org/oclc/1117727465.rdf" />
<link rel="meta" type="application/ld+json" href="http://www.worldcat.org/oclc/1117727465.jsonld" />
<link rel="meta" type="text/turtle" href="http://www.worldcat.org/oclc/1117727465.ttl" />
<link rel="meta" type="text/plain" href="http://www.worldcat.org/oclc/1117727465.nt" />
<meta name="keywords" content="Anthropology Philosophy, Anthropology Methodology, Example, Values, Anthrop
```

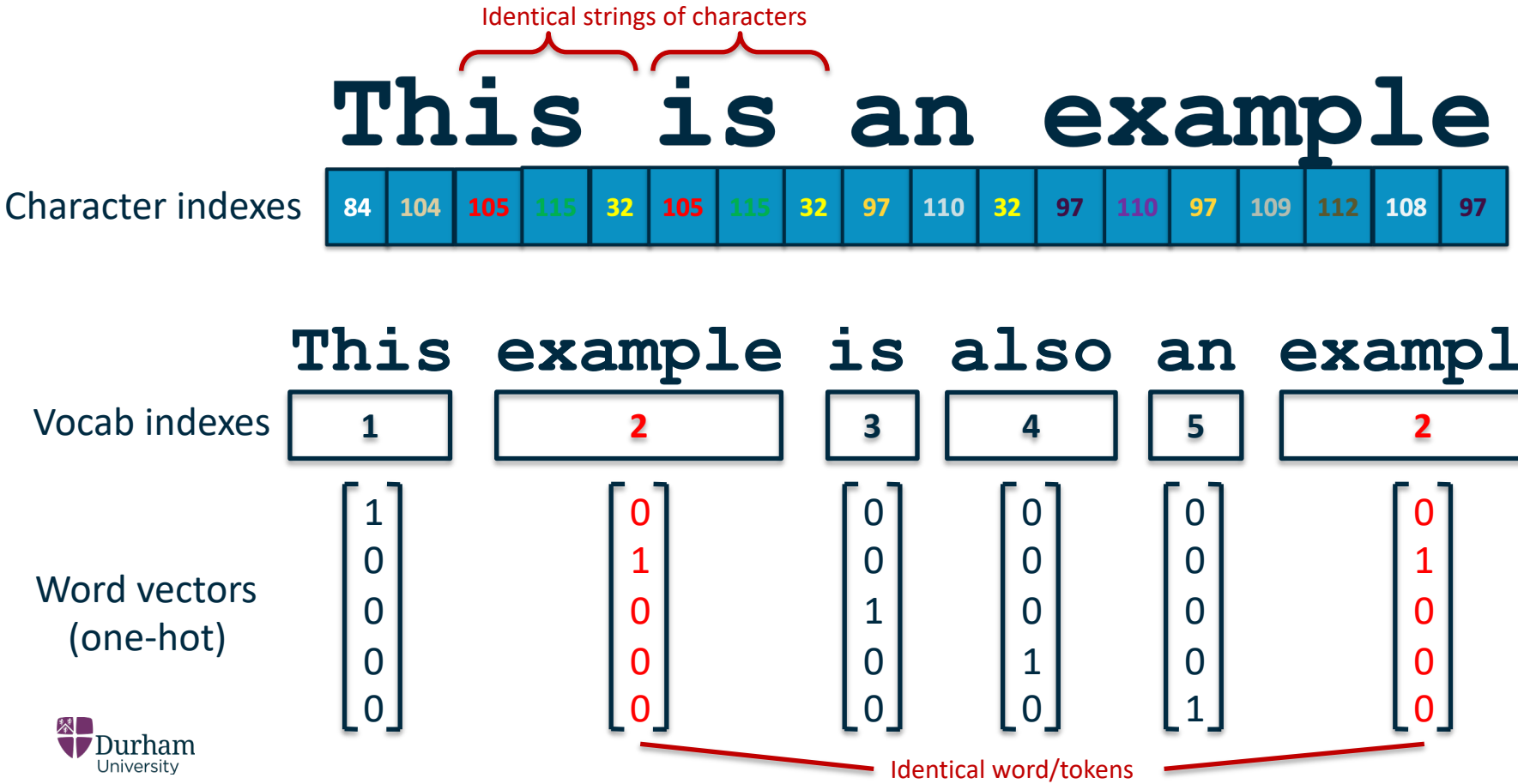
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Internationalization done right: Unicode

- Unicode:
  - Defines what *codepoints* there are (i.e. vocabulary)
  - Assigns a unique ID to each codepoint and defines its semantics
    - E.g. "A" = 65 "萬" = 33836 "😊" = 128522
  - Is *not* an encoding [when people talk like it is, they usually mean UTF-8]
- UTF-8:
  - (One) encoding that can *encode* any Unicode codepoint (serialization)
  - 1:1 *mapping* between byte sequences and codepoints
  - Variable length per codepoint – e.g. "A" = [65] "萬" = [228, 184, 135]
  - Now effectively the "default" encoding (replacing ASCII)

= well-defined method for turning a string of characters into a sequence of bytes

Modelling text



Beyond characters (and codepoints)

S_1 = `The Power of Example: Anthropological Explorations in Persuasion, Evocation.`

S_2 = `The power of example : anthropological explorations in persuasion, evocation, and imitation`

S_3 = `An artificial example about an ant, a rat, and a powerful person`

Q: How similar are these titles in terms of *content*?

- For short strings (e.g. titles), we *could* align and compare the strings*
- For longer / very different strings (e.g. books), this would not work at all

Intuition: at document scale, *words* will be more useful units than *characters*

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* We will do exactly this later on in this module, when we're interested in a slightly different type of comparison

Beyond characters (and codepoints)

S_1 = `The Power of Example: Anthropological Explorations in Persuasion, Evocation.`

S_2 = `The power of example : anthropological explorations in persuasion, evocation, and imitation`

S_3 = `An artificial example about an ant, a rat, and a powerful person`

Q: Is Power == power ?

Q: Is Example: == example : ?

Beyond characters (and codepoints)

S_1 = ~~The Power of Example: Anthropological Explorations in Persuasion, Evocation.~~

S_2 = The power of example : anthropological explorations in persuasion, evocation, and imitation

S_3 = An artificial example about an ant, a rat, and a powerful person

Q: Is powerful == power ?

Q: Is powerless == power ?

Q: Is powerboat == power ?

…

Information Retrieval (IR)

S_1 = `The Power of Example: Anthropological Explorations in Persuasion, Evocation.`

S_2 = `The power of example : anthropological explorations in persuasion, evocation, and imitation`

S_3 = `An artificial example about an ant, a rat, and a powerful person`

Approach 1

- Represent each word (token) by a unique number (vocab index)
- Treat tokens as equivalent iff numbers identical
- Represent each string (document) by a list of numbers (vocab indexes)

Tokenization

S_1 = `The Power of Example: Anthropological Explorations in Persuasion, Evocation.`

D_1 = `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`

		Index
V = [	The	0
	Power	1
	of	2
	Example	3
	:	4
	Anthropological	5
	Explorations	6
	in	7
	Persuasion	8
	,	9
	Evocation	10
	.]	11

Tokenization is a language-specific task!

- Many alphabetic languages (e.g. English): words are punctuation-delimited
- Language-specific normalization (e.g. A==a ?)
  - Simple approach: lowercase all content first

Tokenization

S_1 = the power of example: anthropological explorations in persuasion, evocation.

D_1 = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]

S_2 = the power of example : anthropological explorations in persuasion, evocation, and imitation

D_2 = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 9, 12, 13]

S_3 = an artificial example about an ant, a rat, and a powerful person

D_3 = [14, 15, 3, 16, 14, 17, 9, 18, 19, 9, 12, 18, 20, 21]

V = [

	Index
the	0
power	1
of	2
example	3
:	4
anthropological	5
explorations	6
in	7
persuasion	8
,	9
evocation	10
.	11
and	12
imitation	13
an	14
artificial	15
about	16
ant	17
a	18
rat	19
powerful	20
person]	21

Tokenization

S_1 = 保持共產黨員先進性教育活動

T_1 =

保持	共產黨員	先進性	教育	活動
uphold	Communist party members	advanced	education	campaign

T_2 =

保持	共產黨員	先進	性教育	活動
uphold	Communist party members	advanced	sex education	campaign

- Both syntactically valid tokenizations
- Different meanings
- No common rules across all languages
- Tokenization for e.g. Chinese is a non-trivial NLP task

Language specific task!

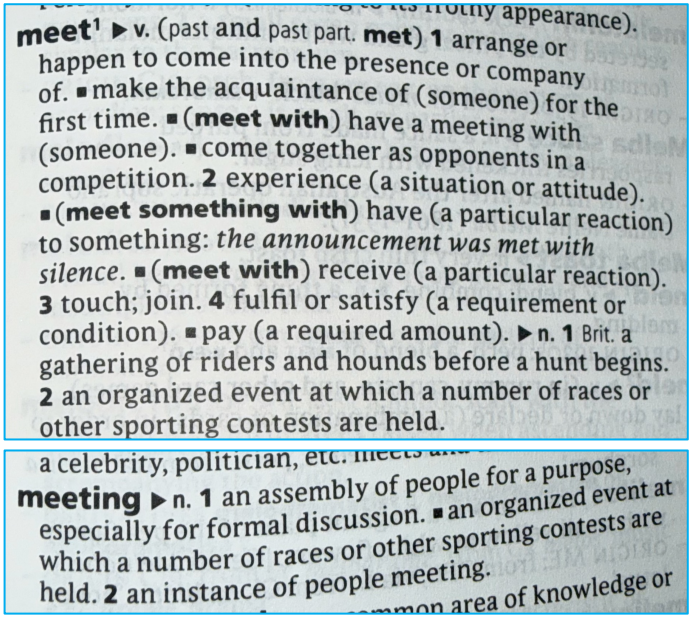
- Not trivial for all languages
  - Not necessarily as trivial as it seems even for *alphabetic* languages
  - E.g. do we really want “New York” as two tokens “New” and “York”?

Normalization: Lemmatization

I hate boring meetings ⇒ meeting
today's meeting ⇒ meeting
I'm meeting friends ⇒ (to) meet
He met a sticky end ⇒ (to) meet

Replace every word with its dictionary form/headword

- bicycle => **bicycle**
- bicycles => **bicycle**
- bicycled => **bicycle**
- bicycling => **bicycle**
- operations => **operation**
- operational => **operation**
- strategic => **strategy**
- strategy => **strategy**



meet[1] ▶ v. (past and past part. **met**) **1** arrange or happen to come into the presence or company of. ▪ make the acquaintance of (someone) for the first time. ▪ (**meet with**) have a meeting with (someone). ▪ come together as opponents in a competition. **2** experience (a situation or attitude). ▪ (**meet something with**) have (a particular reaction) to something: *the announcement was met with silence*. ▪ (**meet with**) receive (a particular reaction). **3** touch; join. **4** fulfil or satisfy (a requirement or condition). ▪ pay (a required amount). ▶ n. **1** Brit. a gathering of riders and hounds before a hunt begins. **2** an organized event at which a number of races or other sporting contests are held.

meeting ▶ n. **1** an assembly of people for a purpose, especially for formal discussion. ▪ an organized event at which a number of races or other sporting contests are held. **2** an instance of people meeting.



bicker verb (bickers, bickering, bickered) argue about unimportant things.
bicycle noun a two-wheeled vehicle that you ride by pushing the pedals with your feet. • verb (bicycles, bicycling, bicycled) ride a bicycle.
bid[1] verb (bids, bidding, bid) **1** offer a price for something. **2** (bid for) offer to do work for a stated price. **3** (bid for) try to get. • noun an act of bidding.
bid[2] verb (bids, bidding, bid or bade; past participle bid) **1** say a greeting. **2** old use command.
biddable adjective obedient.
biddy noun (plural biddies) informal an old woman.
bide verb (bides, biding, bided) old use or dialect stay in a place. □ **bide your time** wait patiently for an opportunity to do something.

Normalization: Lemmatization

Replace every word with its dictionary form/headword

- bicycle => **bicycle**
- bicycles => **bicycle**
- bicycled => **bicycle**
- bicycling => **bicycle**
- operations => **operation**
- operational => **operation**
- strategic => **strategy**
- strategy => **strategy**

For example compressed and compression are both accepted as equivalent to compress



For example compress and compress be both accept as equivalent to compress

Normalization: Stemming

I hate boring meetings ⇒ meet
today's meeting ⇒ meet
I'm meeting friends ⇒ meet
He met a sticky end ⇒ met

"Hacky" algorithms that unify multiple forms of a word (for a given language).

E.g. Porter's algorithm:

Step 1a

sses → ss	*caresses → caress*
ies → i	*ponies → poni*
ss → ss	*caress → caress*
s → ø	*cats → cat*

Step 1b (iff v is a vowel)

(*v*)ing → ø	*walking → walk*
(*v*)ed → ø	*plastered → plaster*

Step 1c (iff v is a vowel)

(*v*)y → i	*pony → poni*

Step 2 (long stems)

ational → ate	*relational→ relate*
izer → ize	*digitizer → digitize*
ator → ate	*operator → operate*

Step 3 (longer stems)

al → ø	*revival → reviv*
able → ø	*adjustable → adjust*
ate → ø	*activate → activ*

...

For example compressed and compression are both accepted as equivalent to compress



For **exampl compress** and **compress ar** both **accept** as **equival** to compress

Normalization: Stop words

Some words may not appear to contribute much *content* to our text

- E.g. suppose we asked: “what are the key concepts in this text?”
  - and or is of in at are not …
    - Likely to be some of the most frequent words!
    - Could never (?) tell us much about *concepts*

Stop words: words we choose to remove entirely

- This film was great => film great
- The film was not great => film great

For example compressed and compression are both accepted as equivalent to compress



example compressed compression both accepted equivalent compress

N-grams

≈ words

N-gram is a sequence of N tokens (for some N ∈ {1, 2, 3, …})

- Capture aspects of observed word usage in a corpus
  - Useful e.g. for language modelling
  - Useful for text reuse identification
- Very straightforward to compute (for small N)

≈ (often big) bunch of text we're interested in

N-gram frequency: how often an N-gram occurs (or: proportion of text)

For small N, generally not possible to reconstruct the original corpus

- Important e.g. for copyright: *corpus* cannot be distributed, N-grams can be

N-grams: simple example

Alice was beginning to get very tired of sitting by her sister on the bank, and of having nothing to do: once or twice she had peeped into the book her sister was reading, but it had no pictures or conversations in it, 'and what is the use of a book,' thought Alice 'without pictures or conversations?'

So she was considering in her own mind (as well as she could, for the hot day made her feel very sleepy and stupid), whether the pleasure of making a daisy-chain would be worth the trouble of getting up and picking the daisies, when suddenly a White Rabbit with pink eyes ran close by her.

Document (or corpus)

1-grams

the	7
of	5
her	5
and	4
was	3
or	3
she	3
a	3
Alice	2
to	2
very	2
by	2

2-grams

by her	2
her sister	2
pictures or	2
or conversations	2
Alice was	1
was beginning	1
beginning to	1
to get	1
get very	1
very tired	1
tired of	1
of sitting	1

3-grams

pictures or conversations	2
Alice was beginning	1
was beginning to	1
beginning to get	1
to get very	1
get very tired	1
very tired of	1
tired of sitting	1
of sitting by	1
sitting by her	1
by her sister	1
her sister on	1

Google n-grams
Building the model

1. Scan 5,195,769 published books from university libraries (“Google Books”)
   - According to Michel, this is “~4% of all books ever published”
2. Run Optical Character Recognition (OCR) on every page of every book
3. Calculate all n-grams (n ∈ {1,2,3,4,5}), and their frequencies in each book
4. Use metadata (language and publication year T) as facets
   - Including publication dates extracted from the OCR
5. Calculate proportion of all writing at year T consisting of each n-gram

“*Quantitative analysis of culture using millions of digitized books*” (Michel et al, 2011, *Science*)
https://books.google.com/ngrams

Google n-grams
Building the model

- Not truly comprehensive, but *a lot* of writing

> The corpus cannot be read by a human. If you tried to read only the entries from the year 2000 alone, at the reasonable pace of 200 words/minute, without interruptions for food or sleep, it would take eighty years. The sequence of letters is one thousand times longer than the human genome: if you wrote it out in a straight line, it would reach to the moon and back 10 times over (8).

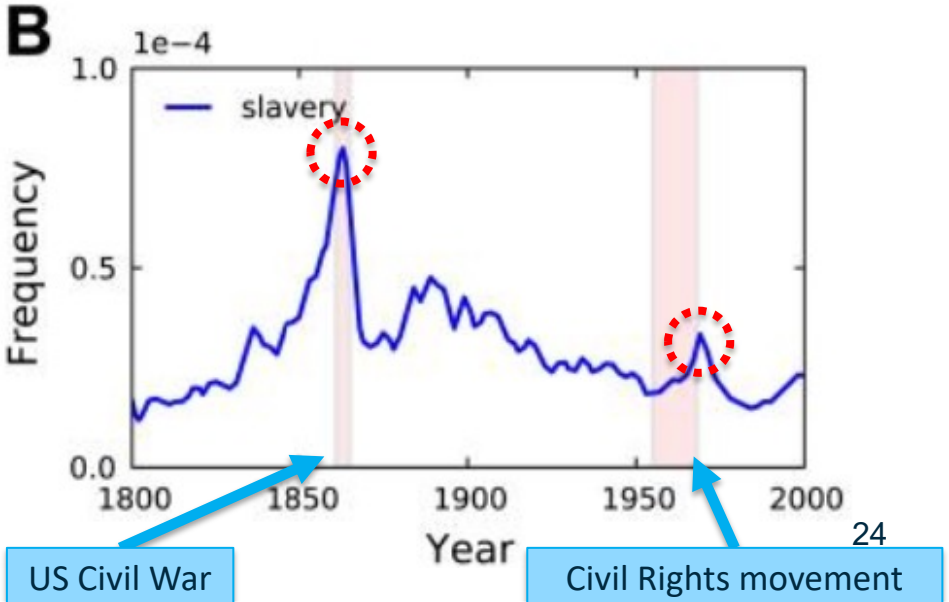
- Frequent criticisms of (some) computational modelling in the humanities:
  - "I could have just *read* those sources and told you that!"
  - "We already *knew* that!"

Google n-grams
Inferences from the model

Effectively, an n-gram language model for every time slice (i.e. each year)

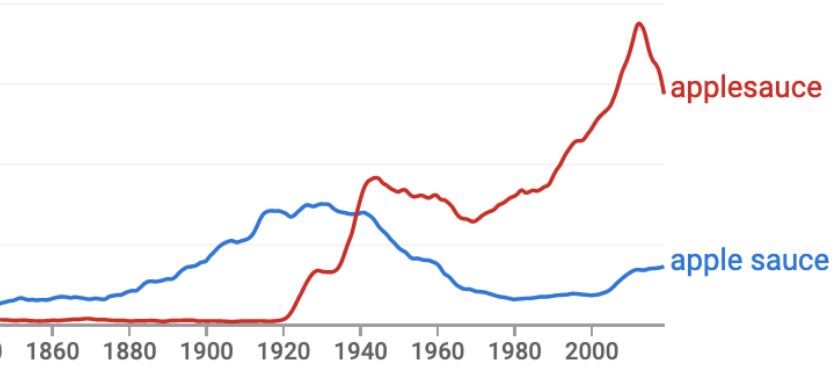
E.g. “how probable is the sequence ‘apple / sauce’ at each t_i”

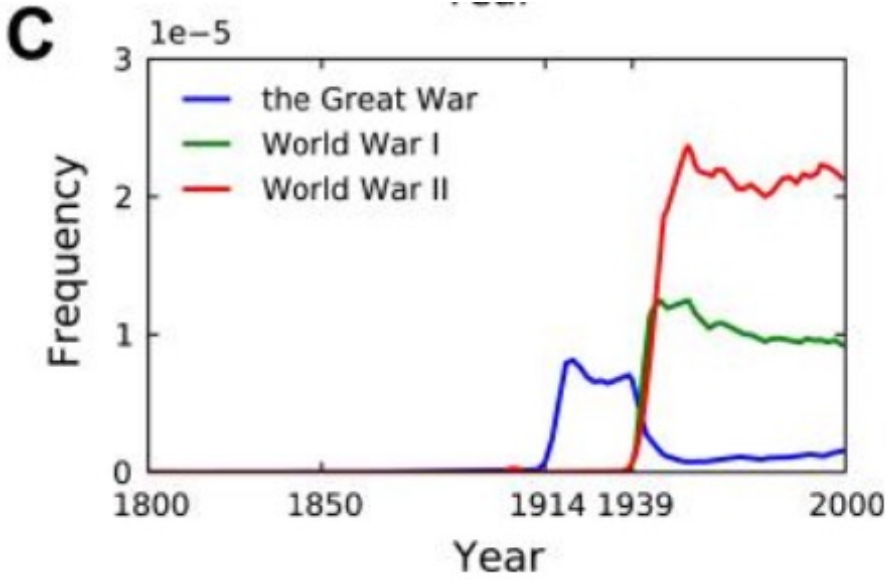
- Word usage representing social trends / societal concerns
  - E.g. “slavery”
  - “what *proportion* of all books published in 1900 consisted of the term ‘slavery’?”



Google n-grams
Inferences from the model

- Word usage reflecting changing *terminology*





Inference from models

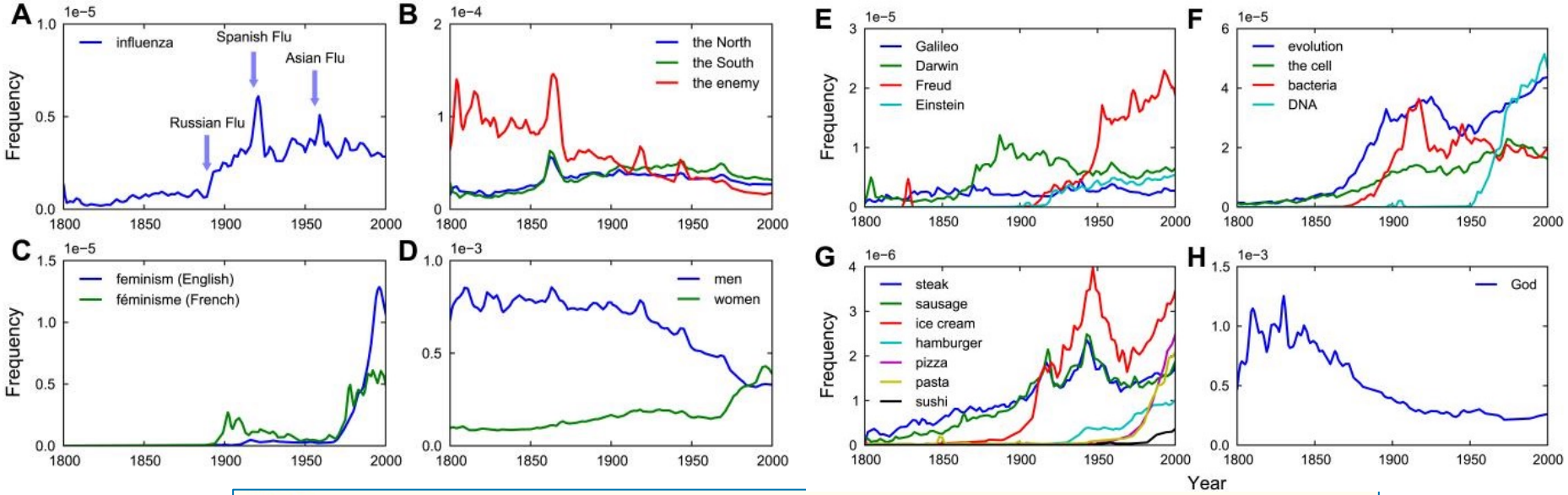


Fig. 5

Culturomics provides quantitative evidence for scholars in many fields. (**A**) Historical Epidemiology: 'influenza' is shown in blue; the Russian, Spanish, and Asian flu epidemics are highlighted. (**B**) History of the Civil War. (**C**) Comparative History. (**D**) Gender studies. (**E,F**) History of Science. (**G**) Historical Gastronomy. (**H**) History of Religion: 'God'.

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Google n-grams
Inferences from the model

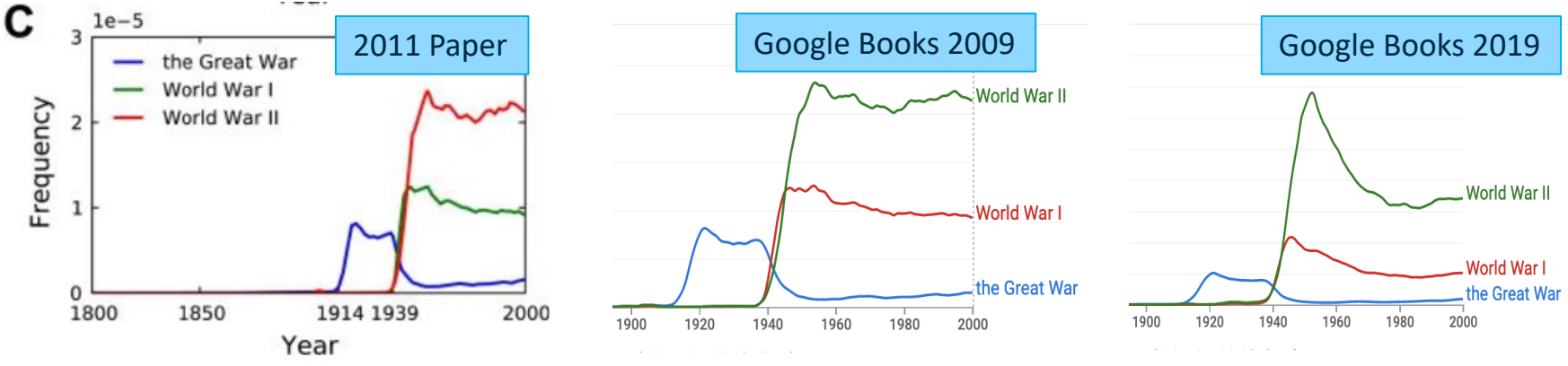
A lot of the *justification* comes from comparison with *intuition*

I.e. charts can be interpreted in ways that seem highly plausible

- "The Great War" became known as "World War I" after World War II

Intuitions confirm *broad* patterns, but *not* detailed data

- Quite a weak standard of confirmation



Google n-grams
Assumptions in drawing inferences from the model

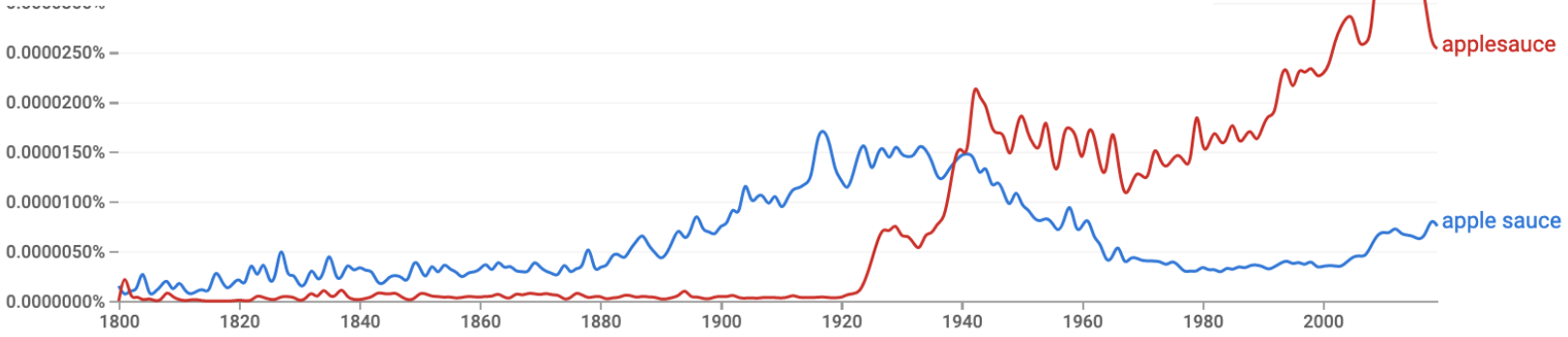
Key assumptions:

1. The books themselves give a representative sample of writing
2. The transcribed text accurately represents the contents of the books
3. The books are assigned approximately accurate publication dates
4. Their contents are representative of writing at the publication date
5. A representative sample of writing approximates cultural trends/concerns
6. A representative sample of writing approximates contemporary language use

- Many criticisms of the project (and use of data) target these assumptions

Inference from models

- Bias
  - E.g. selection of works
- Noise (perhaps [intentionally] hidden)
  - Not necessarily random – e.g. OCR errors
- Insufficient data
- Other explanatory factors present



Google n-grams
Criticism 1: problems with the underlying data

The data contains errors:

a) Scanning errors

b) Metadata errors

c) OCR errors

- All of these might average out, *if* they are random and unbiased

Q: Why/how might each of these be biased?

Q: Why/how might this be a problem?

Take a few moments to think about this

Google n-grams
Criticism 1: problems with the underlying data

Book published 1572 (according to Google)



E. S.

Ake one good turne for an other. Repoſcere mutuũ vel vicem. *Demander qu'on nous rende la pareille.*
Ake an ſue for an office, that another doth the ſame Competo, tis, tij, tere. *Demander la meſme choſe qu'vng autre demande.*
Ake the commons conſent in the partiament houſe. Vide Demaunde, &c.
Ake leaue to ſpeake and to declare his mynde. Dicendi vel agendæ cauſæ locum & poteſtatem petere. *Demander audience.*
Ake and deſire inſtantly to haue iuſtice. Iuſticiæ ſidem iuriſque implorare. *Inſtamment demander*

paruenir a telles louanges.
Aſpiration, or drawyng of brethe. Aſpiratio onis. f. g. *Aſpiration, ſouffiement.*
Aſpire, or to blowe. Aſpiro, as. *Venter, ſouffler.*
Aſſaye or proue. Appetio, iſ. Experior, ris. Tento, as. Facere periculum, Innitor, eris, ixus ſum. Pergo, is, xi. Periclitor, aris. Prætento, as. Probo, as. Tendo, is. *Eſſayer.*
Aſſaye firſte, to take the firſte or former taſte. Prægusto, as. Tendo, is. *Taſter.*
Aſſaye, experience, or proufe. Experientia, Periclitatio. Tentamentum, ti. Tentatio, onis. *Experience.*

Screenshot

APPLE JELLY

5# ripe tart apples or approximately 15 medium apples
5 cups water
9 cups sugar
1 box (1 3/4 oz.) powdered pectin

Wash apples, cutting out blemishes. Cut in small pieces without peeling or coring. The peel, seed, and core are high in pectin, which helps jelly 'jell." Can also add other apple peelings.

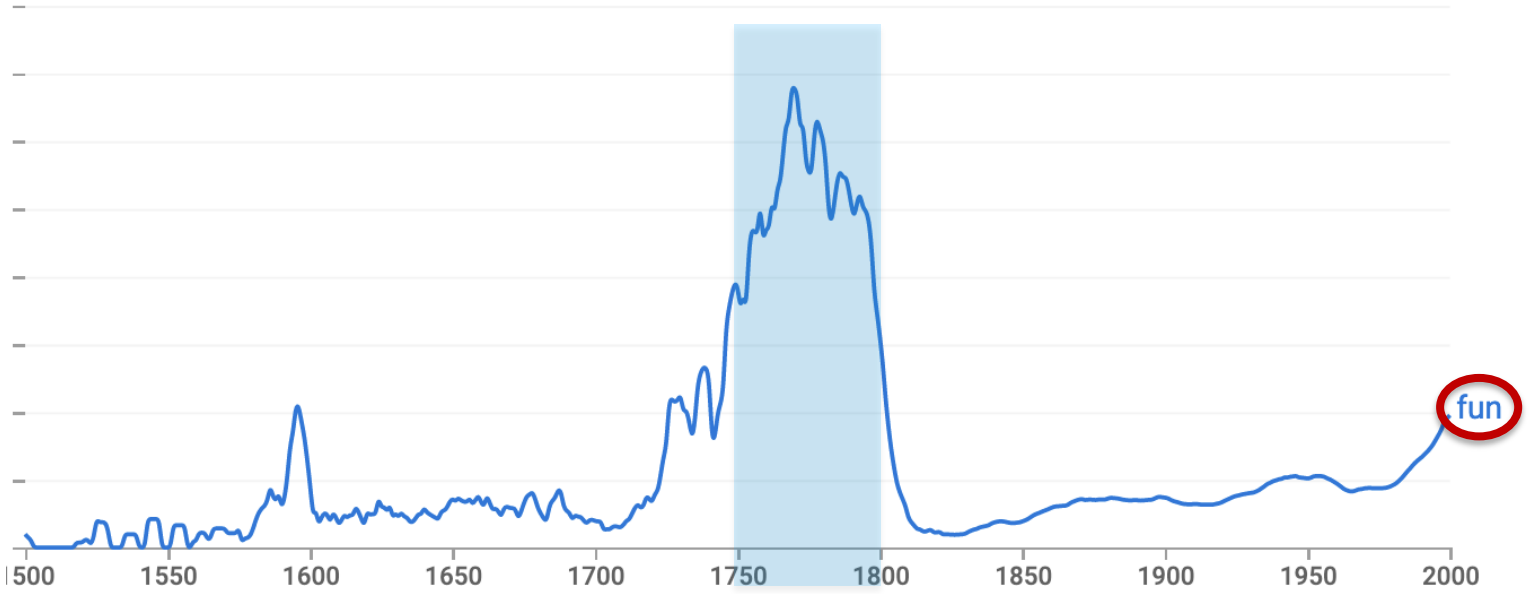
In large pan, cover apples with the water and boil gently, covered, for approxi

Book published 1984

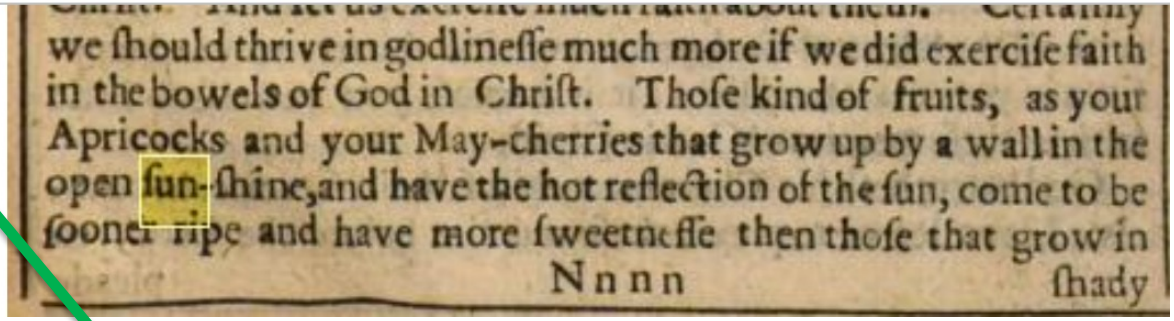


“Peak fun” – 1750-1800AD

Legitimate inference?



Orthographic changes => huge and systematic bias


we ſhould thrive in godlineſſe much more if we did exerciſe faith
in the bowels of God in Chriſt. Thoſe kind of fruits, as your
Apricocks and your May-cherries that grow up by a wall in the
open ſun-ſhine, and have the hot reflection of the ſun, come to be
ſooner ripe and have more ſweetneſſe then thoſe that grow in
Nnnn
ſhady

open ſun-ſhine, and have the hot reflection of the ſun

open sun-shine, and have the hot reflection of the sun

open fun-fhine, and have the hot reflection of the fun

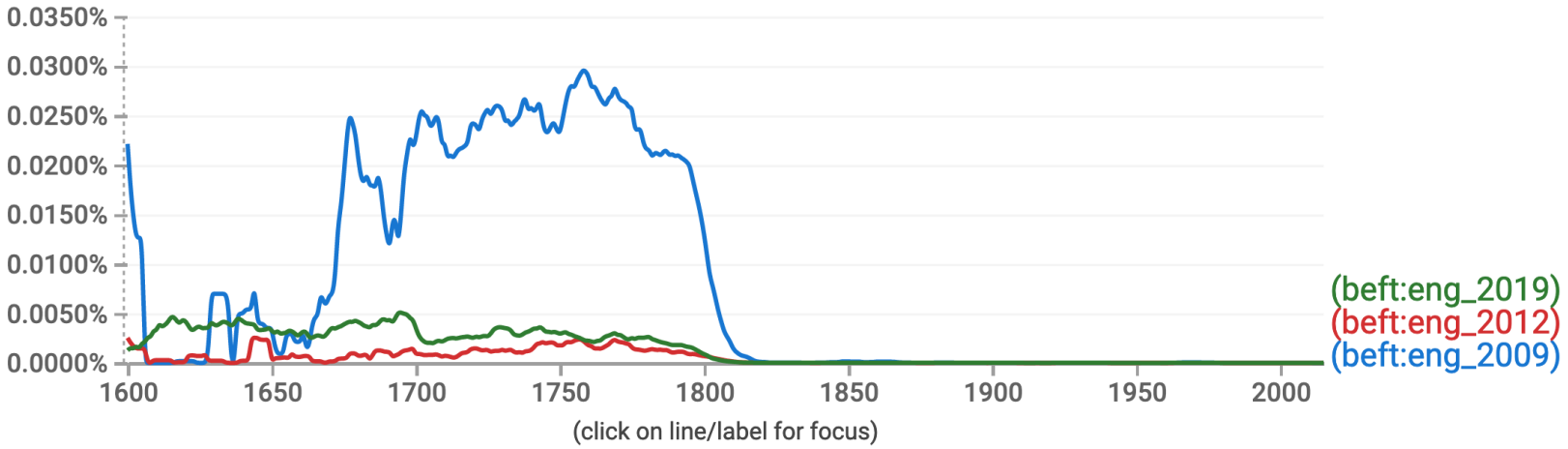
Original text
Modern equivalent
Incorrect OCR

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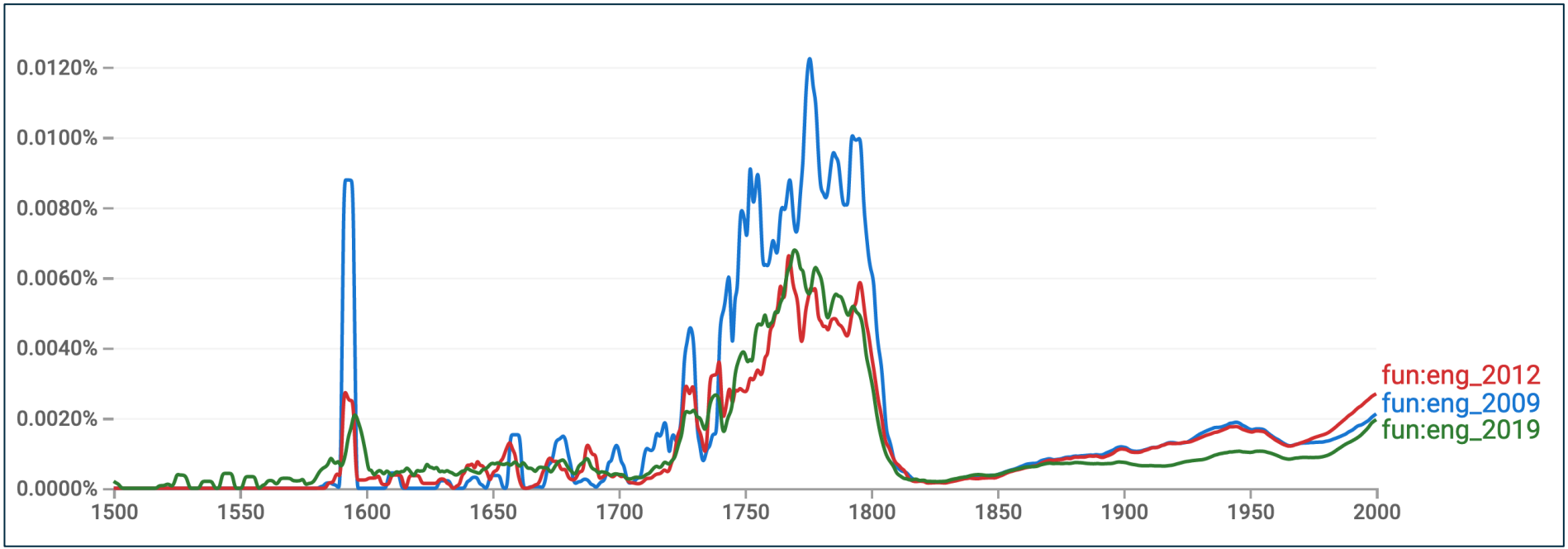
* Identifying other potentially amusing consequences of this effect is left as an exercise to the reader 😉

Google’s response

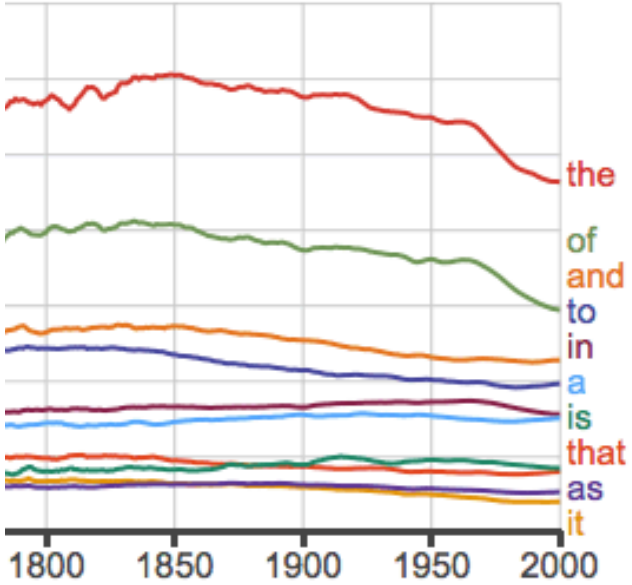
When we generated the original Ngram Viewer corpora in 2009, our OCR wasn't as good as it is today. This was especially obvious in pre-19th century English, where the elongated medial-s (ſ) was often interpreted as an *f*, so *best* was often read as *beft*. Here's evidence of the improvements we've made since then, using the corpus operator to compare the 2009, 2012 and 2019 versions:



Success?


0.0120%
0.0100%
0.0080%
0.0060%
0.0040%
0.0020%
0.0000%
1500
1550
1600
1650
1700
1750
1800
1850
1900
1950
2000
fun:eng_2012
fun:eng_2009
fun:eng_2019

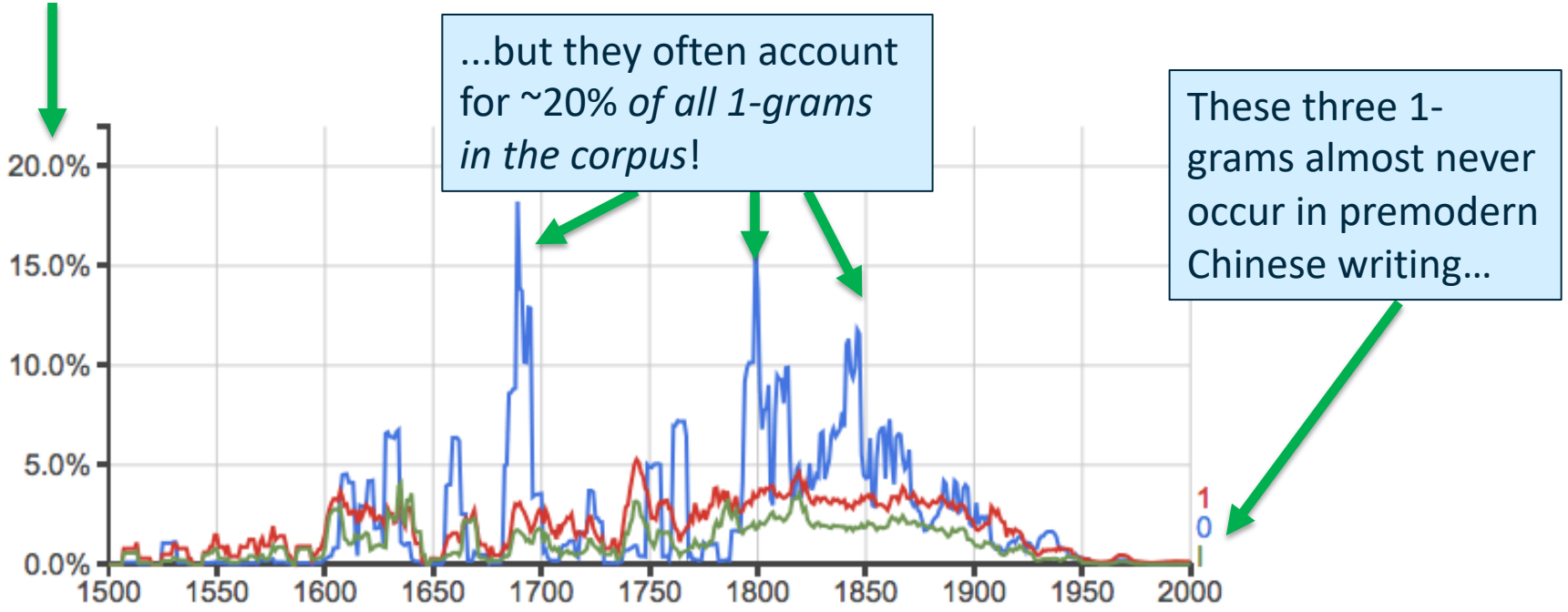
Insufficient data to draw *any* inferences for early T



Ten most frequent 1-grams in English

Normalization isn't trivial

Normalization – i.e. dividing n-gram counts by total words *within each year*



Data for the Google Books Chinese corpus

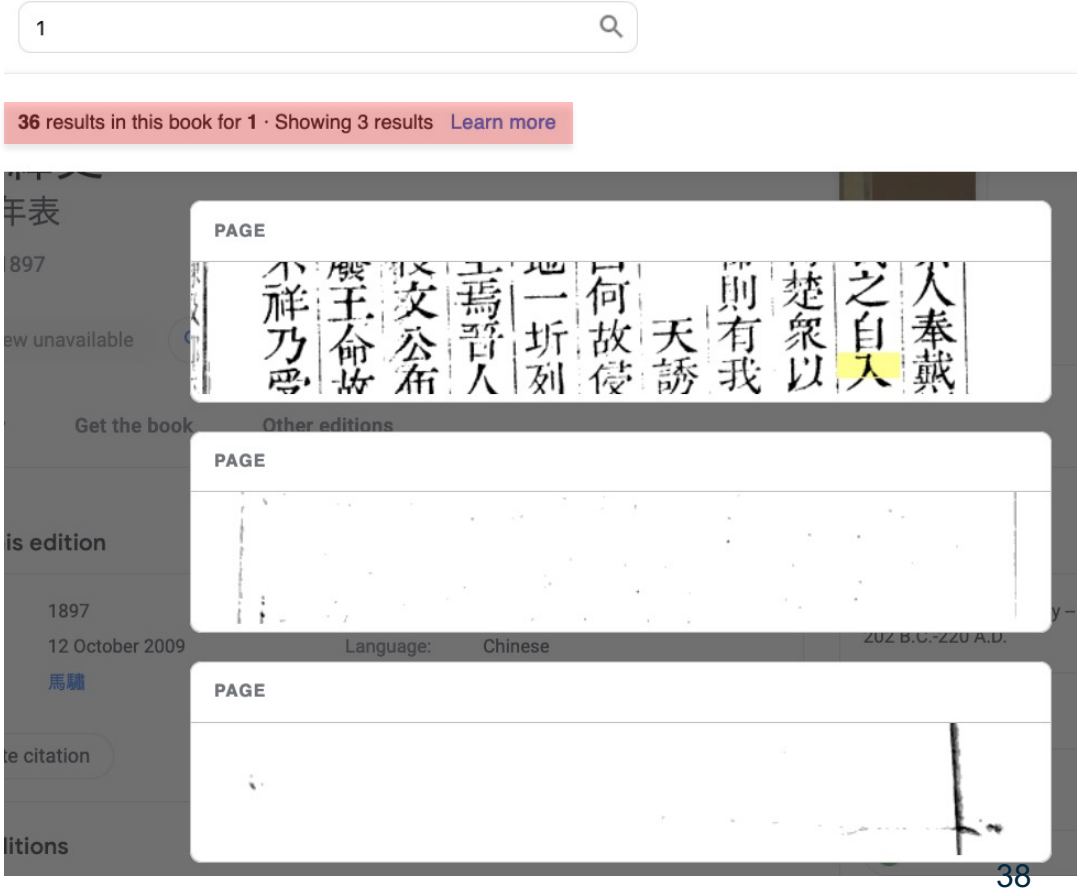
Normalization isn't trivial

Poor noise rejection

⇒ Bias in normalization (e.g. *recent* works easier to OCR)

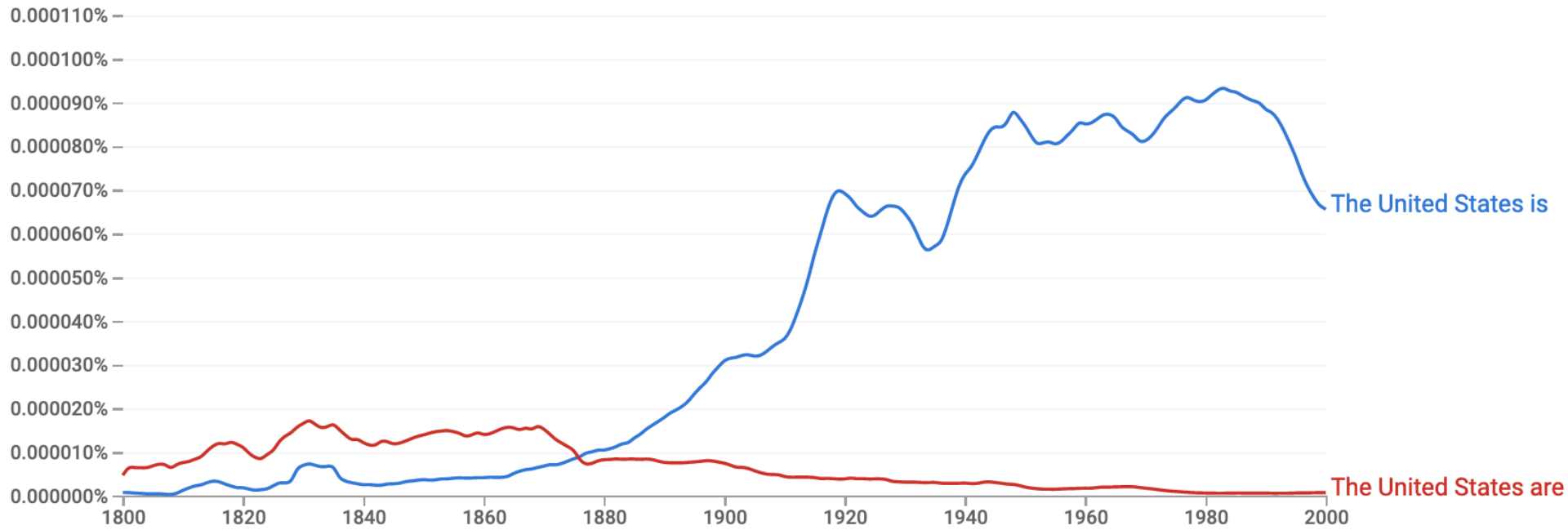
⇒ *More* mistakes where there is *less* data

1、|11」」1,1;,|君王其圍不殼不救荷復 4 ^之躰王弗忍欲許&簏遮諫日臣簡之聖入之劝特&1 吳王帥其賢良真^重緣以上姑蘇使王樣难行成恭趙日背者上天降禍恭夾将皋於亦^:之顙也君^曰 1 #殘女杜喪^汝 5 ^寡人^灭^何面 0 以現^天下乎趨君其夫也速^ ...
第 99 頁
人,^之^^三^^王 0 魏 141 一巾尺^;:-:卜—^:寡人^使八 40 干^^八寡人飯;與之&也三^之&^^^王之遇 —一一一小與 1 尬攻其. 1^11^113^3^一」^》曰與食於此者—钶～也」^^腸陽爲楚伐# ^軍殺將得八城移兵而攻齊陬轸爲齊王使晃胳陽存拜賀戰勝# ...
第 112 頁
小人也安年紫聞之故爲酒而^^幼曰舉何以得葬於先生故常見舂於朝^切 0 :巧 I 狗^^0 ^丈夫之相與^曰田單之愛人瘗乃王之救洚也^动常恶『&11 :丧-牛化者復見王曰王^朝:::宜^田一單而楫之於庭口勞|秀布今求百胜之,「力收,^一.寡人之葸單王嘉之单之



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Normalization could be done differently



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Avoiding reliance on normalization

$C_t(x)$ = # of n-grams in corpus at time t that are x

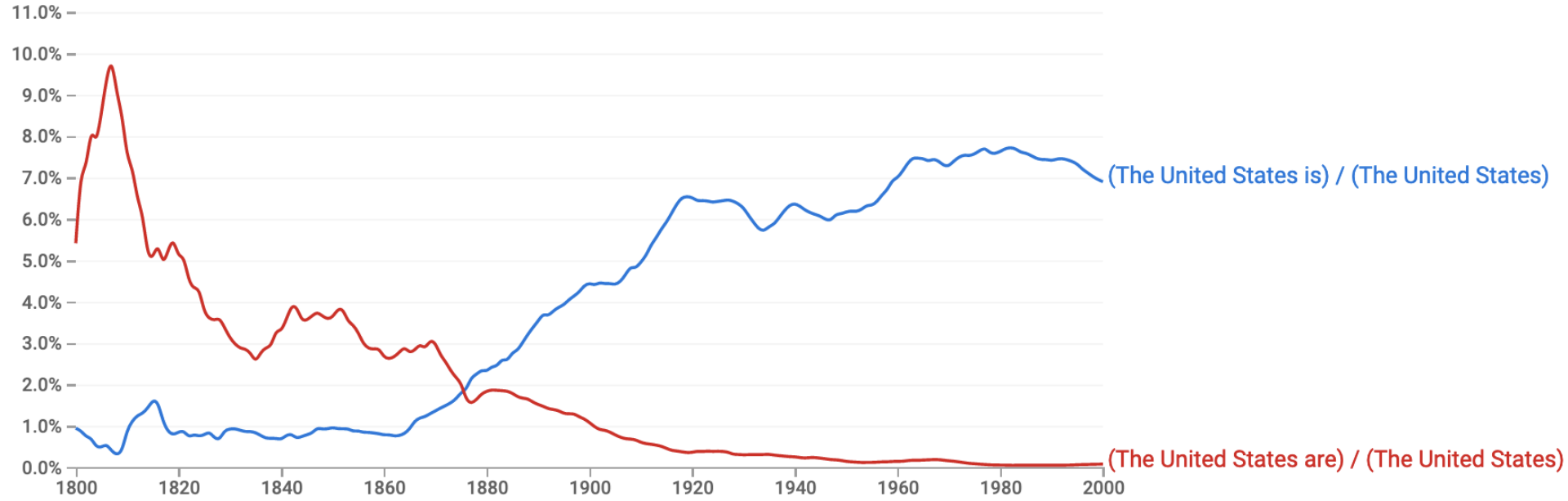
$P_t(x)$ = proportion of corpus at time t that is x

S_t = total corpus size at time t

$$P_t(x) = \frac{C_t(x)}{S_t}$$

$$\frac{P_t(a)}{P_t(b)} = \frac{C_t(a)}{C_t(b)}$$

(Independent of S_t)



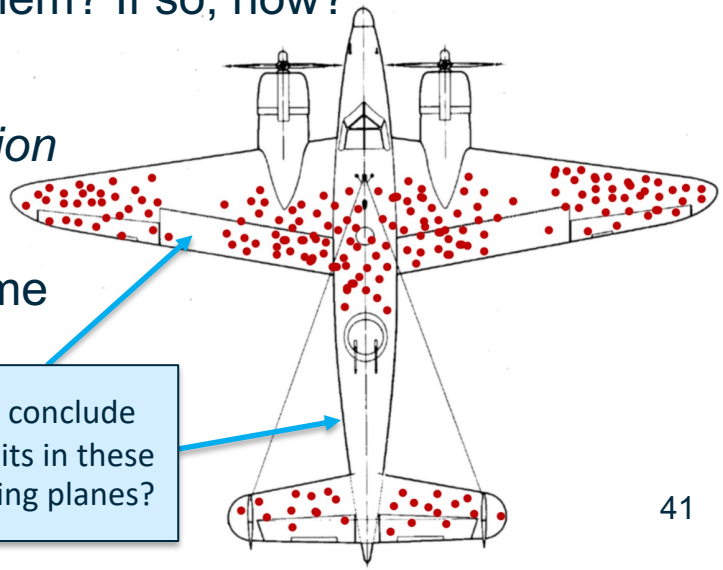
Criticism 2: Is the model built on a fair sample?

Q: What *would be* a fair sample of writing?

- A fair sample of all works published?
  - If I publish 100 nearly identical books, should they all count?
  - Are all books equal, or should we weight them? If so, how?

Survivorship bias?

- Books in *libraries* were selected *for preservation*
- May overrepresent particular *types* of work
  - Proportion of these types may vary over time



Google n-grams
Practical and conceptual challenges!

2 BOOK
Merchant Of Venice
1835
Available Online

3 BOOK
Merchant Of Venice
1819
Available Online

4 BOOK
Merchant of Venice
1838
Available Online

5 BOOK
Merchant of Venice
1890
Available Online

6 BOOK
Merchant of Venice
1852
Available Online

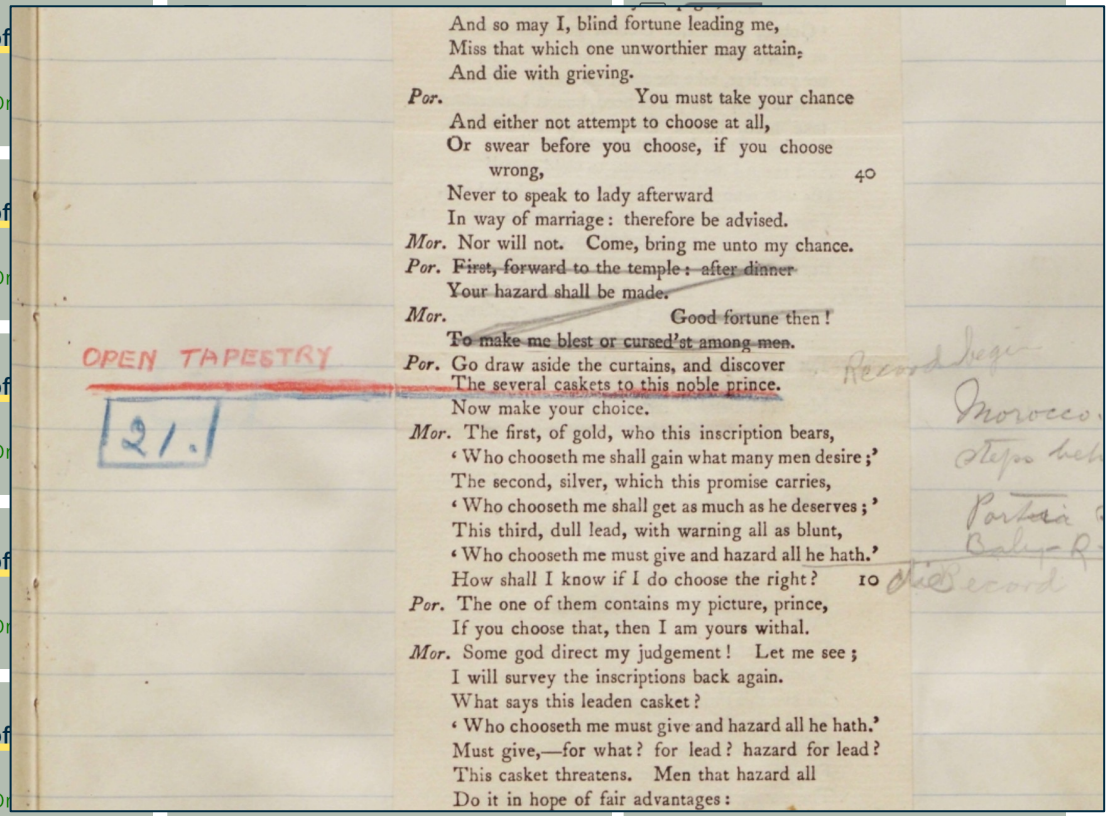
7 BOOK
Merchant of
1780
Available On

8 BOOK
Merchant of
1892
Available On

9 BOOK
Merchant of
1870
Available On

10 BOOK
Merchant of
1879
Available On

11 BOOK
Merchant of
1780
Available On



And so may I, blind fortune leading me,
Miss that which one unworthier may attain,
And die with grieving.

Por. You must take your chance
And either not attempt to choose at all,
Or swear before you choose, if you choose wrong,
Never to speak to lady afterward
In way of marriage: therefore be advised.

Mor. Nor will not. Come, bring me unto my chance.

Por. First, forward to the temple: after dinner
Your hazard shall be made.

Mor. Good fortune then!
To make me blest or cursed'st among men.

Por. Go draw aside the curtains, and discover
The several caskets to this noble prince.
Now make your choice.

Mor. The first, of gold, who this inscription bears,
'Who chooseth me shall gain what many men desire;'
The second, silver, which this promise carries,
'Who chooseth me shall get as much as he deserves;'
This third, dull lead, with warning all as blunt,
'Who chooseth me must give and hazard all he hath.'
How shall I know if I do choose the right?

Por. The one of them contains my picture, prince,
If you choose that, then I am yours withal.

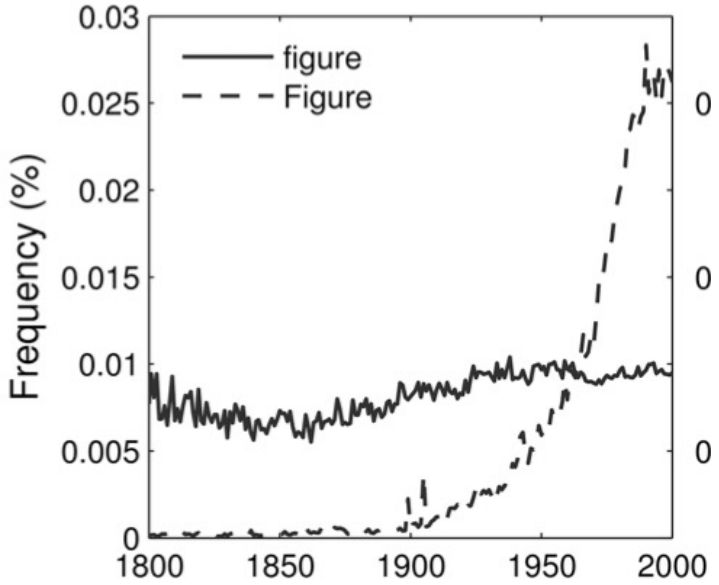
Mor. Some god direct my judgement! Let me see;
I will survey the inscriptions back again.
What says this leaden casket?
'Who chooseth me must give and hazard all he hath.'
Must give,—for what? for lead? hazard for lead?
This casket threatens. Men that hazard all
Do it in hope of fair advantages:

OPEN TAPESTRY
21.

Criticism 2: Is the model built on a fair sample?

Scientific research *overrepresented*

- Influence of library *collection* practices
- Hence these works will be *oversampled*



https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0137041